Controlled Testing and Failure Attribution for ML Sensor Pipelines

Paper 3 proposal

Working proposal for postdoc and advisor review

28 July 2026

Abstract

Post-training monitors flag distribution change but rarely say whether the evidence is most consistent with a sensor/acquisition shift, an unsupported physical state, or a broken evaluation contract. We propose a controlled, replay-based study of that distinction over *declared intervention families*—not unrestricted root-cause discovery, and not a maintained software product. Declared interventions, named representation hooks, deterministic contract gates, and calibrated abstention are evaluated on NuBench point clouds and a fixed waveform reconstruction task, against input tests, MMD, classifier two-sample tests, embedding distance, uncertainty, and output-only monitoring. Because the waveform testbed generates its own noise covariance, we also measure what each shift *costs* the reconstruction in the likelihood geometry of Papers 1–2, and test whether detectability and consequence agree. Intended results: generic monitors detect N/S shifts; layerwise evidence improves the hard N-versus-S separation; deterministic gates isolate evaluation faults; and monitor response is only partially aligned with real reconstruction loss.

1 Scope, continuity, and novelty

Paper 1 identifies likelihood geometry under structured noise; Paper 2 separates residual geometry, architecture class, and training support. Paper 3 asks whether those contracts can be tested after training, and whether a violation that is *detectable* is also *consequential*. Its artifact is a reproducible experiment suite—configurations, adapters, hooks, replay scripts, and reports—not a branded platform or a promise of production maintenance.

The contribution is a controlled cross-modal design that combines (i) paired sensor perturbations, support-composition shifts, and deliberate contract faults; (ii) feature-matched attribution comparisons with guarded abstention; (iii) a detectability-versus-consequence comparison on the arm where the true covariance is known by construction; and (iv) a power–cost study of full capture, sampling, reservoirs, and fixed-dimensional sketches. NuBench, the waveform-noise generator, and MMD are inputs to that design, not claimed inventions. Cross-geometry results test consistency, not causal detector-design effects.

Identifiability limit, from Paper 1. Diagonalizing a covariance yields decorrelated statistical modes, not labeled physical sources: $\Sigma = \Sigma_1 + \Sigma_2$ admits infinitely many PSD-cone splits and low-rank factors are rotation-degenerate. Attribution here is over declared intervention families under a stated simulator; a response profile does not identify the cause of an unknown real shift.

2 Failure contracts and claims

We predeclare three labels. **N—acquisition/noise shift** includes PSD or covariance change, jitter, drift, spectral lines, glitches, tails, hit thinning, and channel/module loss. **S—training-support shift** changes window composition toward physical strata that are sparse under the clean training distribution; events themselves are not corrupted. **E—evaluation-contract fault** includes wrong units, coordinate ordering, output semantics, PSD convention, or metric implementation. E is not a distribution shift and is not the likelihood-metric mismatch studied in Papers 1–2.

N versus S is the discriminative core. E is decided by deterministic schema and range invariants, so it is a gate result (E0), not a class in a classification score. The informative separation is N from S: input-quality evidence identifies N directly, whereas S leaves events uncorrupted and is input-invisible by construction, so it must be detected from representation position relative to the training support. If the layerwise signature cannot separate N from S, C2 has no interesting content regardless of E.

C1—Detect N/S shifts. At a 1% false-alert rate on nominal, non-overlapping windows, the best monitoring signature detects at least 80% of held-out abrupt and gradual N/S shifts within five windows and is non-inferior to the strongest eligible baseline by a five-point power margin. Delay is primary in *windows*; events and simulated seconds are reported per testbed, and cross-modal statements use windows only.

C2—Attribute with abstention. The primary endpoint is macro-F1 over {N, S} on held-out cases, with N+S mixtures scored by multi-label F1 in the same endpoint; E-gate sensitivity/specificity is reported separately under E0. We predeclare *superiority* of the full layerwise signature over the all-generic-monitor arm as a paired difference whose 95% CI excludes zero, with a smallest relevant effect of 0.10 macro-F1 on {N, S}, rather than a fixed point margin: a real effect should not be scored as a failure because a strong baseline performed well. Split-conformal distance in standardized fingerprint space must identify an unseen family with AUROC ≥ 0.80 instead of forcing a known label. This is intervention-family attribution, not unrestricted physical root-cause discovery.

C3—Trade power for cost. Ten-percent event sampling plus fixed-size sketches yields p95 replay-path latency overhead $< 10\%$, at least $10\times$ less representation storage, and no more than five points of power/F1 loss relative to full capture. Effects and intervals are repeated across modalities, three NuBench model families, two machines, and CPU/GPU where supported.

C4—Detection is not consequence. On the waveform arm the true Σ is generated, so each N intervention has a computable cost in the likelihood geometry of Papers 1–2: change in whitened residual, subspace angle against the clean basis, amplitude bias, and calibration ratio. We predeclare that monitor response and consequence are only partially aligned—rank correlation below a threshold fixed at freeze, with both off-diagonal cells of Table 1 populated. This makes the likelihood geometry load-bearing rather than cited: the consequence column cannot be computed without knowing which residual geometry is correct. No analogous quantity exists on NuBench, where consequence is limited to downstream angular error; we state that asymmetry rather than implying symmetric coverage. All targets above are frozen after development profiling and power analysis.

3 Methods and testbeds

Each event window records input-quality summaries, named-layer moments or sketches, outputs, uncertainty, delayed task metrics, and calibration. Eligible N/S baselines receive identical windows and false-alert budgets: corrected univariate input tests, RBF-MMD [2], a classifier two-sample test [3], final-embedding mean/covariance distance, output tests, and uncertainty/entropy. A regularized logistic classifier is fitted with identical splits to five arms: input only; output/uncertainty; final embedding only; all generic-monitor outputs; and the full layerwise signature. The fourth arm is the strong baseline, committed to in advance. E uses schema/range invariants and clean-reference tests, not drift statistics. Clean miscalibration is a model result, not automatically a software bug [4]; latent displacement is normalized by clean within-class dispersion.

NuBench point clouds. The first cell is Hexagon Ice LE $\times$ DynEdge $\times$ direction. The restored 1,358,099-parameter checkpoint exposes a 128-D pre-head event representation. Module dropout, hit thinning, and time jitter instantiate N. S windows use clean low-density strata in energy, topology, vertex/containment, and multiplicity, estimated from training only. ParticleNeT and GRIT repeat the study; another geometry is a consistency check [1].

Waveforms and the noise module. The fixed task uses non-overlapping 4096-sample windows at 10 MS/s and sinusoid-amplitude reconstruction. Two encoders are fitted on the chronological training split: the rank-2 noise-aware EMPCA encoder of Paper 1 and the Kronecker-factored encoder of Paper 2, so the waveform arm is not a single-model testbed. The module accepts or constructs PSD/CSD models, samples seeded frequency-domain random coefficients with the required real-signal symmetry, transforms them into time traces, and adds stationary color, piecewise nonstationarity, lines, drift, glitches, and tails to clean signals (Fig. 1b–c). It validates units and second-order spectral statistics. It is therefore a controlled N intervention engine, not another dataset, reconstruction model, or physical-source separator. S holds out declared frequency/amplitude ranges; E swaps coefficient order or frequency units and changes PSD/score conventions.

Because the module generates the noise, both the assumed $\hat{\Sigma}$ used by the encoder and the realized Σ are known for every intervention, so each waveform window carries a consequence record (C4) alongside its monitor record — one additional column, not a separate experiment.

Table 1: Detectability versus consequence (C4, waveform arm). The off-diagonal cells are the result.

	Low consequence	High consequence
Monitor fires	**False alarm that matters**: priced against C3 cost	True positive
Monitor silent	Correctly ignored	**Silent failure**: the dangerous cell

Table 2: Confirmatory experiment matrix. Drift baselines apply only to N/S.

ID	Test	Primary endpoints
E0	Clean replay; deliberate E faults	Reference discrepancy; gate sensitivity/specificity
E1	Held-out abrupt/gradual N/S	Power at 1% false alert; AUROC/AUPRC; delay in windows; consequence (waveform)
E2	N/S attribution, N+S mixtures, unseen family	Macro/multi-label F1 over {N,S}; paired CI vs generic arm; unknown AUROC; detection–consequence rank correlation
E3	Capture vs sample/reservoir/sketch	p50/p95 latency; bytes/event; power/F1 loss
E4	Models, modalities, hardware	Scaling; interval agreement; report hashes

Held-out severities, seeds, support strata, N+S mixtures, and one unseen family prevent perturbation lookup. Windows and thresholds are frozen after development. Intervals resample event groups and perturbation seeds separately. Cost sweeps event rate, window size, hook count, embedding dimension, and model count; repeat runs record machine, accelerator, timing method, and data/config hashes. Null results require a passed positive control, adequate precision, and a predeclared equivalence margin.

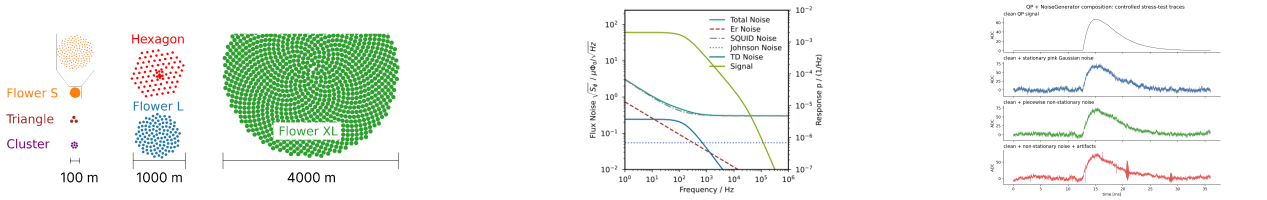
4 Optional Track B: a collaborator pipeline

The study above runs on public NuBench data and our own waveform task; no external dependency is on the critical path. A third testbed on a collaborator's pipeline would test what neither of ours can—whether a standardized diagnostic conclusion *replicates* outside the domains it was developed in. It is offered as an option to accept, scope down, or decline.

Scope if adopted, deliberately small. Only the smallest supported conclusion from E1/E2—one representation, one severity grid, paired pre/post events—reported as replication or heterogeneity. Tracks are never pooled: they share only the contract *paired input* $\rightarrow$ *declared intervention and severity* $\rightarrow$ *frozen representation* $\rightarrow$ *standardized response*. Raw severities and latent distances are never compared across modalities.

Feasibility turns on four answers. (i) Input modality—dense waveforms admit the noise module directly; hits or higher-level objects need a different intervention channel and make the module precedent only. (ii) Availability of raw inputs for paired re-inference. (iii) Which intermediate representation extracts reproducibly. (iv) Which data are real versus simulated, and what noise-state metadata exist.

Three participation levels, to choose from. *(a) Review only*—comment on perturbation plausibility and physical interpretation; no data exchange. *(b) Third testbed*—supply pipeline and representation access for the reduced replication



(a) NuBench detector geometries (b) detector signal/noise spectrum (c) generated waveform stress tests

Figure 1: Complementary testbeds. (a) Six top-view geometries, reproduced from NuBench Fig. 3 [1]. (b) The supplied Al_2O_3/Al athermal-detector response and component PSDs. (c) The generator composes a clean optimal-filter-like pulse with stationary colored noise, piecewise nonstationarity, and explicit artifacts. Panels (b)–(c) are project outputs.

above. *(c) Joint consequence study*—co-develop C4 as a systematic-uncertainty analysis, where a surrogate or fast forward model's approximation error is exactly a declared N intervention and its measured consequence is what a systematics budget requires. Level (c) is the closest fit to a physics-analysis interest and the only one that changes the claim set.

5 Smoke test and feasibility gate

The selected 8.82 GiB NuBench release is checksummed; the DynEdge checkpoint restores and the 128-D hook runs. Rescoring all 2,989,339 released predictions reproduces published median angular errors within 0.055°: 23.6548° versus 23.60° for CC/track and 56.1553° versus 56.12° for NC/cascade. This misses a deliberately strict 0.01° identity tolerance. On 256 development events, checkpoint and released directions differ by 0.944° median (7.708° p95); therefore the clean gate remains open.

Both discrepancies are high in the same direction and of comparable size (+0.055°, +0.035°) on two independent populations. Rescoring released predictions is deterministic, so this is more consistent with an event-selection or metric-definition difference—a quality cut, a different denominator, or median-of-medians versus pooled median—than with backend numerics. We test selection variants before negotiating a tolerance, noting that 0.01° is 0.042% of a 23.65° median. A module-dropout positive control is monotone (15.28° → 25.12° median angular error; 10-NN retention 1.000 → 0.393 from 0–50% dropout), establishing perturbation, hook, and paired-metric feasibility without presenting the pilot as a scientific claim.

References

[1] R. F. Ørsøe et al. "NuBench: An Open Benchmark for Deep Learning-Based Event Reconstruction in Neutrino Telescopes." JINST 21 (2026). arXiv:2511.13111.

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[3] D. Lopez-Paz and M. Oquab. "Revisiting Classifier Two-Sample Tests." ICLR (2017).

[4] C. Guo et al. "On Calibration of Modern Neural Networks." ICML (2017).